

Tasks:

1. Read the .csv file using Pandas. Take a look at the top few records.
2. Normalize casings for the review text and extract the text into a list for easier manipulation.
3. Tokenize the reviews using NLTKs word_tokenize function.
4. Perform parts-of-speech tagging on each sentence using the NLTK POS tagger.
5. For the topic model, we should want to include only nouns.
 1. Find out all the POS tags that correspond to nouns.
 2. Limit the data to only terms with these tags.
6. Lemmatize.
 1. Different forms of the terms need to be treated as one.
 2. No need to provide POS tag to lemmatizer for now.
7. Remove stopwords and punctuation (if there are any).
8. Create a topic model using LDA on the cleaned-up data with 12 topics.
 1. Print out the top terms for each topic.
 2. What is the coherence of the model with the c_v metric?
9. Analyze the topics through the business lens.
 1. Determine which of the topics can be combined.
10. Create topic model using LDA with what you think is the optimal number of topics
 1. What is the coherence of the model?
11. The business should be able to interpret the topics.
 1. Name each of the identified topics.
 - b. Create a table with the topic name and the top 10 terms in each to present to the business.